

Functional enrichment of species-specific gene families in *Trypodendron lineatum*

Response to Reviewer 1, comment #4 (lines 446-448)

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4/24/26

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1 Aim

Functional enrichment analysis (GO + Pfam domains) of the species-specific gene families of *T. lineatum* identified by OrthoFinder.

Definition of species-specific: orthogroups where only *T. lineatum* has genes assigned (all other species empty). This excludes singletons (genes in `Orthogroups_UnassignedGenes.tsv`) and captures expanded gene families exclusive to the *T. lineatum* lineage relative to the 8 species compared.

Note on namespace conflicts. Loading `topGO` pulls in `AnnotationDbi`, `IRanges` and `Biobase`, which mask several base/tidyverse functions (`select`, `filter`, `setdiff`, `intersect`, ...). Throughout this script we use explicit `package::function` prefixes so the code runs correctly in a clean R session, independently of the order in which libraries are loaded.

2 1. Setup

```
# ---- Installation (uncomment only the first time) ----
# install.packages(c("BiocManager", "tidyverse", "data.table",
#                   "ggplot2", "scales"))
# BiocManager::install(c("topGO", "Rgraphviz"))

suppressPackageStartupMessages({
  library(tidyverse)
  library(data.table)
  library(topGO)
  library(scales)
})

set.seed(42)

out_dir <- file.path("results", "enrichment_species_specific")
dir.create(out_dir, recursive = TRUE, showWarnings = FALSE)
```

3 2. Parameters

```
# Update these paths to wherever your files live
interpro_tsv <- "Tlin_interpro.tsv"
orthogroups_tsv <- "Orthogroups.tsv"

# EXACT name of the T. lineatum column in Orthogroups.tsv
target_species <- "Fix_Trypodendron_lineatum"

# Thresholds
fdr_threshold <- 0.05
min_node_size <- 5      # topGO: minimum genes annotated per GO term
min_pfam_count <- 2     # Pfam: domains with at least N genes to be tested
```

4 3. Parsing InterProScan

InterProScan TSV with `--goterms`. The file has 13 or 14 columns depending on the row (column 14 = GO terms when there is an InterPro match; absent otherwise). We use `data.table::fread` for efficient, memory-safe parsing of the ragged TSV.

```
ipr_cols <- c("protein_id", "md5", "length", "analysis",
             "sig_acc", "sig_desc", "start", "end",
             "score", "status", "date",
             "ipr_acc", "ipr_desc", "go_terms", "pathways")

ipr_dt <- data.table::fread(interpro_tsv, sep = "\t", header = FALSE,
                          fill = TRUE, na.strings = c("", "NA", "-"),
                          colClasses = "character", quote = "")

data.table::setnames(ipr_dt, old = names(ipr_dt),
                    new = ipr_cols[seq_len(ncol(ipr_dt))])

ipr <- tibble::as_tibble(ipr_dt)
rm(ipr_dt); invisible(gc())

cat("Unique proteins in IPS:", dplyr::n_distinct(ipr$protein_id), "\n")
```

Unique proteins in IPS: 10512

```
cat("Analyses present:      ", paste(unique(ipr$analysis), collapse = ", "), "\n")
```

Analyses present: PANTHER, Pfam

```
cat("Total rows:          ", nrow(ipr), "\n")
```

Total rows: 40119

4.1 3.1 gene → GO mapping

```
gene2go_df <- ipr %>%
  dplyr::filter(!is.na(go_terms)) %>%
  dplyr::select(protein_id, go_terms) %>%
  tidyr::separate_rows(go_terms, sep = "\\|") %>%
  dplyr::mutate(go_terms = stringr::str_extract(go_terms, "GO:\\d{7}")) %>%
  dplyr::filter(!is.na(go_terms)) %>%
```

```

dplyr::distinct()

# Named list for topGO: gene -> vector of GO IDs
gene2GO <- split(gene2go_df$go_terms, gene2go_df$protein_id)

cat("Genes with at least one GO: ", length(gene2GO), "\n")

```

Genes with at least one GO: 6703

```

cat("Total gene-GO assignments: ", nrow(gene2go_df), "\n")

```

Total gene-GO assignments: 15520

4.2 3.2 gene → Pfam mapping

```

gene2pfam <- ipr %>%
  dplyr::filter(analysis == "Pfam") %>%
  dplyr::select(protein_id, sig_acc, sig_desc) %>%
  dplyr::distinct()

cat("Genes with at least one Pfam:", dplyr::n_distinct(gene2pfam$protein_id), "\n")

```

Genes with at least one Pfam: 9503

```

cat("Unique Pfam domains: ", dplyr::n_distinct(gene2pfam$sig_acc), "\n")

```

Unique Pfam domains: 4456

5 4. Identifying species-specific orthogroups

```

og <- readr::read_tsv(orthogroups_tsv, col_types = readr::cols(.default = "c"))

stopifnot(target_species %in% colnames(og))

other_species <- base::setdiff(colnames(og), c("Orthogroup", target_species))

og_specific <- og %>%
  dplyr::filter(!is.na(.data[[target_species]]) & .data[[target_species]] != "") %>%
  dplyr::filter(dplyr::if_all(dplyr::all_of(other_species), ~ is.na(.) | . == ""))

```

```
cat("Species-specific orthogroups:", nrow(og_specific), "\n")
```

Species-specific orthogroups: 73

```
specific_genes <- og_specific %>%
  dplyr::pull(!target_species) %>%
  stringr::str_split(",\\s*") %>%
  unlist() %>%
  stringr::str_trim() %>%
  purrr::discard(~ .x == "") %>%
  unique()

cat("Genes in species-specific orthogroups:", length(specific_genes), "\n")
```

Genes in species-specific orthogroups: 228

```
cat(" - With InterProScan annotation:      ",
    sum(specific_genes %in% ipr$protein_id), "\n")
```

- With InterProScan annotation: 42

```
cat(" - With at least one GO term:          ",
    sum(specific_genes %in% names(gene2GO)), "\n")
```

- With at least one GO term: 17

```
# Save for Supplementary Material
writeLines(specific_genes,
           file.path(out_dir, "S_TableX_species_specific_gene_ids.txt"))
readr::write_tsv(og_specific,
                 file.path(out_dir, "S_TableX_species_specific_orthogroups.tsv"))
```

6 5. GO enrichment with topGO

6.1 5.1 Universe

The universe is all *T. lineatum* genes with at least one GO assignment from InterProScan. Genes without any GO term contribute no information to the test.

```
all_annotated <- names(gene2GO)
gene_factor <- factor(as.integer(all_annotated %in% specific_genes))
names(gene_factor) <- all_annotated

cat("Universe (genes with GO):", length(gene_factor), "\n")
```

```
Universe (genes with GO): 6703
```

```
cat("Test set (specific genes with GO): ", sum(gene_factor == 1), "\n")
```

```
Test set (specific genes with GO): 17
```

```
cat("Test set / Universe (%):",
    round(100 * sum(gene_factor == 1) / length(gene_factor), 2), "%\n")
```

```
Test set / Universe (%): 0.25 %
```

6.2 5.2 Enrichment function

`weight01` (topGO's recommended default) corrects for the GO hierarchy by discounting signal that propagates from child to parent terms. We also report `classic` (uncorrected) and `BH`-adjusted FDR for transparency.

```
run_topgo <- function(ontology) {
  go_data <- new("topGOdata",
    description = paste("Species-specific", ontology),
    ontology    = ontology,
    allGenes    = gene_factor,
    annot       = annFUN.gene2GO,
    gene2GO     = gene2GO,
    nodeSize    = min_node_size)

  result_w01 <- runTest(go_data, algorithm = "weight01", statistic = "fisher")
  result_cla <- runTest(go_data, algorithm = "classic",  statistic = "fisher")

  n_nodes <- length(usedGO(go_data))

  tab <- GenTable(go_data,
    classicFisher = result_cla,
    weightFisher  = result_w01,
    orderBy       = "weightFisher",
    topNodes      = n_nodes,
    numChar       = 1000)

  tab <- tab %>%
```

```

dplyr::mutate(
  Ontology      = ontology,
  classicFisher = as.numeric(gsub("<", "", classicFisher)),
  weightFisher  = as.numeric(gsub("<", "", weightFisher)),
  FDR_BH        = p.adjust(weightFisher, method = "BH"),
  fold_enrichment = (Significant / sum(gene_factor == 1)) /
                    (Annotated   / length(gene_factor))
) %>%
tibble::as_tibble()

list(data = go_data, result = result_w01, table = tab)
}

go_bp <- run_topgo("BP")
go_mf <- run_topgo("MF")
go_cc <- run_topgo("CC")

go_all <- dplyr::bind_rows(go_bp$table, go_mf$table, go_cc$table) %>%
  dplyr::arrange(Ontology, weightFisher)

go_sig <- go_all %>% dplyr::filter(weightFisher < fdr_threshold)

cat("Significant GO terms (weightFisher <", fdr_threshold, "):\n")

```

Significant GO terms (weightFisher < 0.05):

```
go_sig %>% dplyr::count(Ontology) %>% print()
```

```

# A tibble: 3 x 2
  Ontology      n
  <chr>      <int>
1 BP          6
2 CC          2
3 MF          6

```

6.3 5.3 Export full table

```

readr::write_tsv(go_all, file.path(out_dir, "S_TableX_GO_enrichment_all.tsv"))
readr::write_tsv(go_sig, file.path(out_dir, "S_TableX_GO_enrichment_significant.tsv"))

```

7 6. Pfam domain enrichment

Fisher's exact test (one-sided, greater) per domain + BH correction.

```

pfam_universe <- unique(gene2pfam$protein_id)
pfam_test_set <- base::intersect(specific_genes, pfam_universe)

n_test <- length(pfam_test_set)
n_univ <- length(pfam_universe)

cat("Pfam universe:", n_univ, "| Pfam test set:", n_test, "\n")

```

Pfam universe: 9503 | Pfam test set: 34

```

pfam_counts <- gene2pfam %>%
  dplyr::group_by(sig_acc, sig_desc) %>%
  dplyr::summarise(
    n_total = dplyr::n_distinct(protein_id),
    n_in_test = sum(unique(protein_id) %in% pfam_test_set),
    .groups = "drop"
  ) %>%
  dplyr::filter(n_total >= min_pfam_count)

pfam_enrich <- pfam_counts %>%
  dplyr::rowwise() %>%
  dplyr::mutate(
    a = n_in_test, # test + with domain
    b = n_total - n_in_test, # non-test + with domain
    c = n_test - n_in_test, # test + no domain
    d = (n_univ - n_test) - (n_total - n_in_test), # non-test + no domain
    pvalue = fisher.test(matrix(c(a, c, b, d), nrow = 2),
      alternative = "greater")$p.value,
    fold_enrichment = (n_in_test / n_test) / (n_total / n_univ)
  ) %>%
  dplyr::ungroup() %>%
  dplyr::mutate(FDR = p.adjust(pvalue, method = "BH")) %>%
  dplyr::arrange(FDR, pvalue) %>%
  dplyr::select(sig_acc, sig_desc,
    n_in_test, n_total, fold_enrichment, pvalue, FDR,
    dplyr::everything(), -a, -b, -c, -d)

pfam_sig <- pfam_enrich %>% dplyr::filter(FDR < fdr_threshold, n_in_test >= 2)

cat("Significant Pfam domains (FDR <", fdr_threshold, "):", nrow(pfam_sig), "\n")

```

Significant Pfam domains (FDR < 0.05): 5

```

readr::write_tsv(pfam_enrich, file.path(out_dir, "S_TableX_Pfam_enrichment_all.tsv"))
readr::write_tsv(pfam_sig, file.path(out_dir, "S_TableX_Pfam_enrichment_significant.tsv"))

```


8 7. Visualization

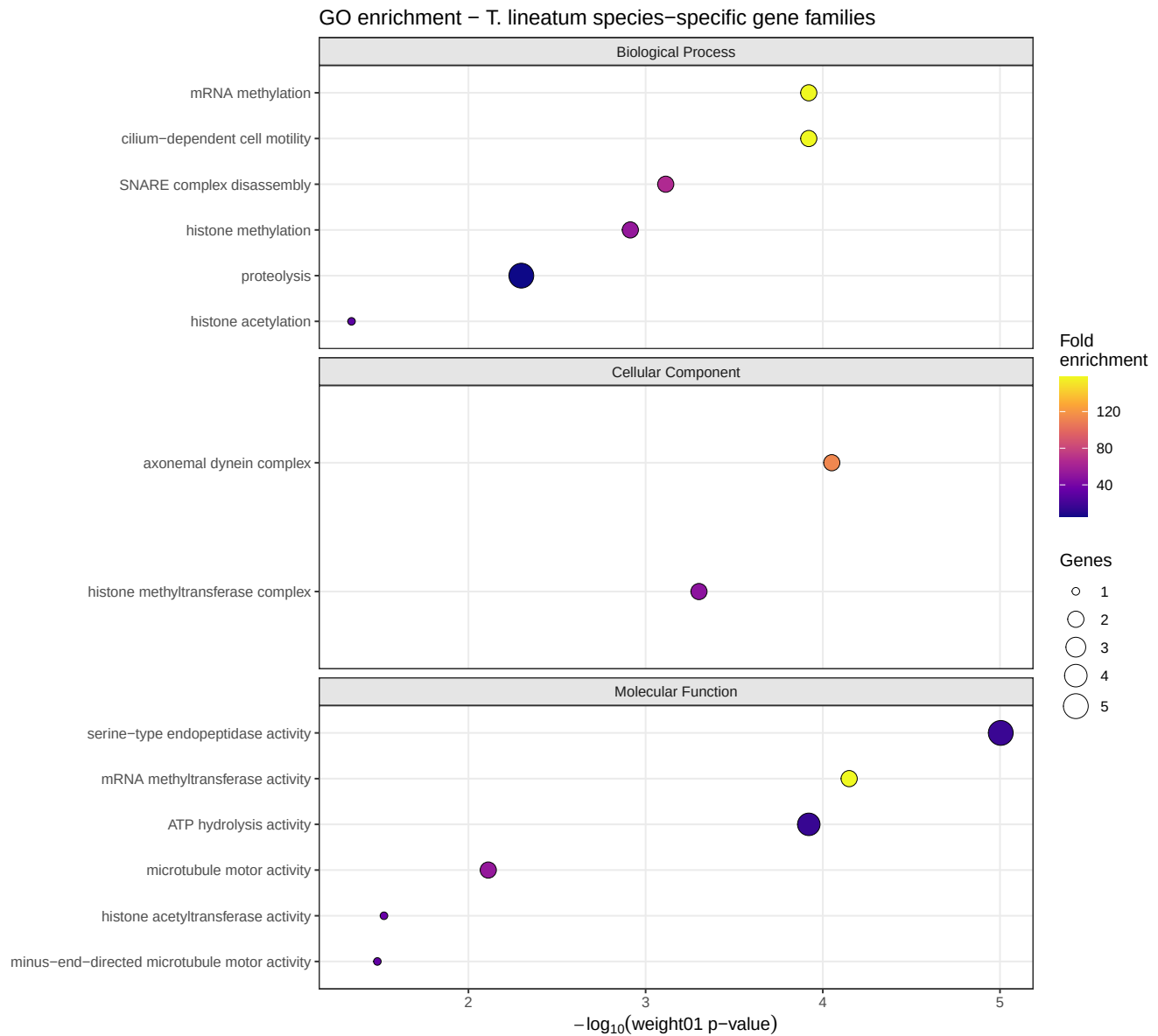
8.1 7.1 GO: top terms per ontology

```
if (nrow(go_sig) > 0) {
  plot_go <- go_sig %>%
    dplyr::group_by(Ontology) %>%
    dplyr::slice_min(weightFisher, n = 15) %>%
    dplyr::ungroup() %>%
    dplyr::mutate(
      Term = stringr::str_trunc(Term, 55),
      Term = forcats::fct_reorder(Term, -log10(weightFisher))
    )

  p_go <- ggplot2::ggplot(plot_go,
    ggplot2::aes(x = -log10(weightFisher), y = Term,
      fill = fold_enrichment, size = Significant)) +
    ggplot2::geom_point(shape = 21, color = "black", stroke = 0.3) +
    ggplot2::facet_wrap(~ Ontology, scales = "free_y", ncol = 1,
      labeller = ggplot2::labeller(Ontology = c(
        BP = "Biological Process",
        MF = "Molecular Function",
        CC = "Cellular Component")))) +
    ggplot2::scale_fill_viridis_c(option = "plasma", name = "Fold\\enrichment") +
    ggplot2::scale_size_continuous(name = "Genes", range = c(2, 7)) +
    ggplot2::labs(x = expression(-log[10]("weight01 p-value")), y = NULL,
      title = "GO enrichment - T. lineatum species-specific gene families") +
    ggplot2::theme_bw(base_size = 11) +
    ggplot2::theme(strip.background = ggplot2::element_rect(fill = "grey90"),
      panel.grid.minor = ggplot2::element_blank())

  print(p_go)

  ggplot2::ggsave(file.path(out_dir, "FigureSX_GO_enrichment.png"),
    p_go, width = 10, height = 9, dpi = 300)
  ggplot2::ggsave(file.path(out_dir, "FigureSX_GO_enrichment.pdf"),
    p_go, width = 10, height = 9)
} else {
  cat("No significant GO terms to plot.\n")
}
```



8.2 7.2 Pfam: top enriched domains

```
if (nrow(pfam_sig) > 0) {
  plot_pfam <- pfam_sig %>%
    dplyr::slice_min(FDR, n = 20) %>%
    dplyr::mutate(
      label = paste0(sig_acc, " - ", stringr::str_trunc(sig_desc, 45)),
      label = forcats::fct_reorder(label, -log10(FDR))
    )

  p_pfam <- ggplot2::ggplot(plot_pfam,
    ggplot2::aes(x = -log10(FDR), y = label,
      fill = fold_enrichment, size = n_in_test)) +
    ggplot2::geom_point(shape = 21, color = "black", stroke = 0.3) +
```

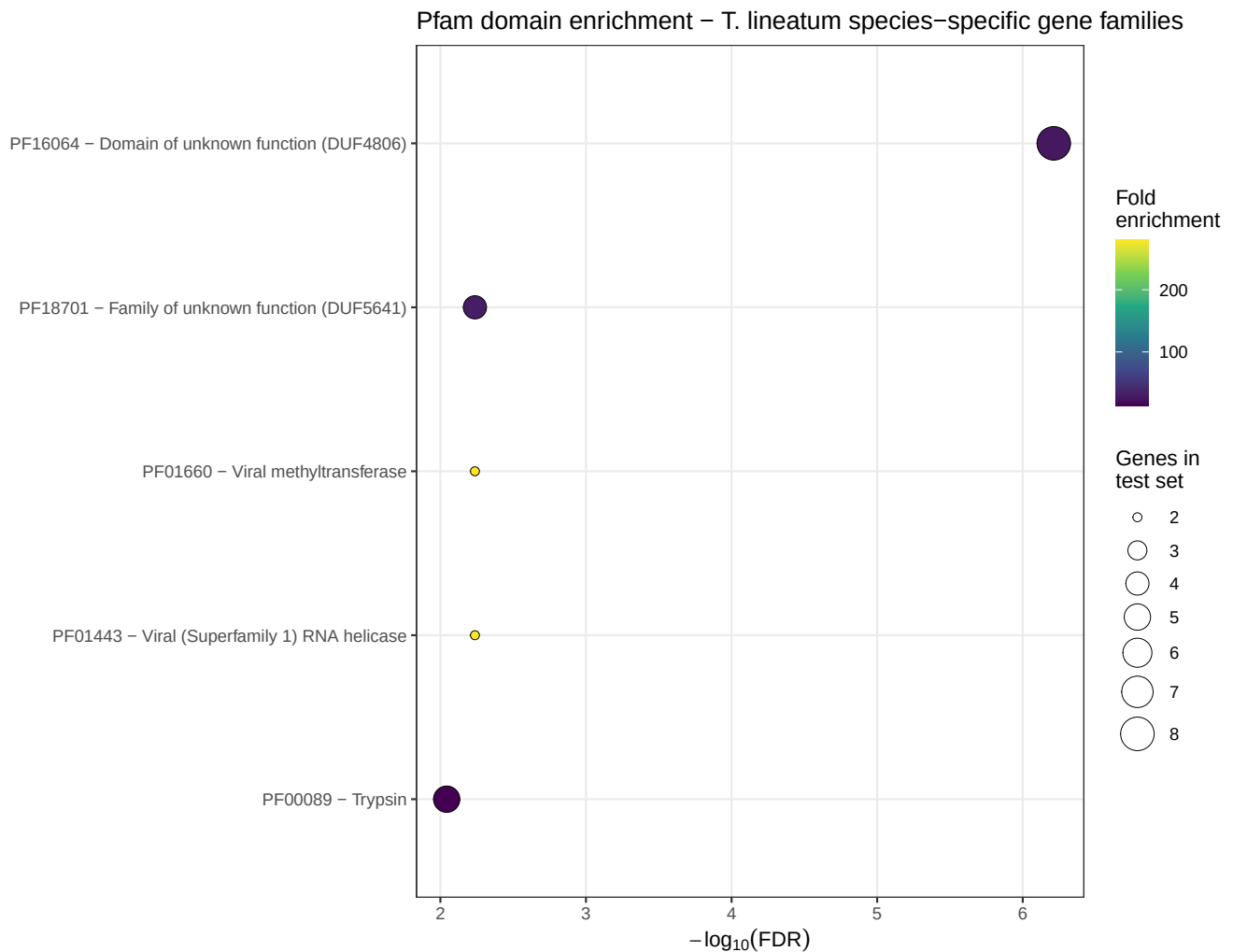
```

ggplot2::scale_fill_viridis_c(option = "viridis", name = "Fold\nenrichment") +
ggplot2::scale_size_continuous(name = "Genes in\ntest set", range = c(2, 8)) +
ggplot2::labs(x = expression(-log[10]("FDR")), y = NULL,
              title = "Pfam domain enrichment - T. lineatum species-specific gene families")
ggplot2::theme_bw(base_size = 11) +
ggplot2::theme(panel.grid.minor = ggplot2::element_blank())

print(p_pfam)

ggplot2::ggsave(file.path(out_dir, "FigureSX_Pfam_enrichment.png"),
                p_pfam, width = 9, height = 7, dpi = 300)
ggplot2::ggsave(file.path(out_dir, "FigureSX_Pfam_enrichment.pdf"),
                p_pfam, width = 9, height = 7)
} else {
  cat("No significant Pfam domains to plot.\n")
}

```



9 8. Summary by biological category

Thematic classification useful for drafting the discussion (lines 446-448).

```
biology_keywords <- list(
  "Detoxification / xenobiotic metabolism" = c("cytochrome P450", "P450",
    "glutathione", "GST", "UDP-glucuronosyl", "carboxylesterase",
    "epoxide", "ABC transporter"),
  "Chemoreception" = c("odorant", "olfactory",
    "gustatory", "chemosensory", "pheromone"),
  "Digestion / carbohydrate metabolism" = c("glycos", "cellulase",
    "xylanase", "mannanase", "chitin", "lipase", "peptidase"),
  "Immunity / defense" = c("immun", "antimicrobial",
    "defensin", "toll", "serine protease inhibitor"),
  "Cuticle / structural" = c("cuticle", "chitin binding"),
  "Reproduction / development" = c("vitellogenin", "yolk",
    "oogenesis", "spermatogenesis")
)

classify_term <- function(term) {
  cats <- purrr::map_chr(names(biology_keywords), function(cat) {
    if (any(stringr::str_detect(tolower(term),
                                tolower(biology_keywords[[cat]])))) cat else NA_character_
  })
  cats <- cats[!is.na(cats)]
  if (length(cats) == 0) "Other / uncharacterized" else paste(cats, collapse = "; ")
}

if (nrow(go_sig) > 0) {
  go_sig_cat <- go_sig %>%
    dplyr::mutate(Category = purrr::map_chr(Term, classify_term)) %>%
    dplyr::arrange(Category, weightFisher)
  readr::write_tsv(go_sig_cat,
    file.path(out_dir, "S_TableX_GO_enrichment_categorized.tsv"))
  cat("=== Significant GO terms by biological category ===\n")
  go_sig_cat %>% dplyr::count(Category, sort = TRUE) %>% print()
}
```

=== Significant GO terms by biological category ===

A tibble: 2 x 2

Category	n
<chr>	<int>
1 Other / uncharacterized	13
2 Digestion / carbohydrate metabolism	1

```
if (nrow(pfam_sig) > 0) {
  pfam_sig_cat <- pfam_sig %>%
    dplyr::mutate(Category = purrr::map_chr(sig_desc, classify_term)) %>%
```

```

    dplyr::arrange(Category, FDR)
  readr::write_tsv(pfam_sig_cat,
                    file.path(out_dir, "S_TableX_Pfam_enrichment_categorized.tsv"))
  cat("\n=== Significant Pfam domains by biological category ===\n")
  pfam_sig_cat %>% dplyr::count(Category, sort = TRUE) %>% print()
}

```

=== Significant Pfam domains by biological category ===

A tibble: 1 x 2

Category	n
<chr>	<int>
1 Other / uncharacterized	5

10 9. Final metrics for the manuscript

```
cat("===== SUMMARY FOR LINES 446-448 =====\n\n")
```

===== SUMMARY FOR LINES 446-448 =====

```
cat("Species-specific orthogroups:      ", nrow(og_specific), "\n")
```

Species-specific orthogroups: 73

```
cat("Tlin genes in these orthogroups:    ", length(specific_genes), "\n")
```

Tlin genes in these orthogroups: 228

```
cat("  With InterProScan annotation:      ",
    sum(specific_genes %in% ipr$protein_id),
    sprintf(" (%.1f%%)\n",
            100 * sum(specific_genes %in% ipr$protein_id) / length(specific_genes)))
```

With InterProScan annotation: 42 (18.4%)

```
cat("  With GO term assigned:              ",
    sum(specific_genes %in% names(gene2GO)),
    sprintf(" (%.1f%%)\n",
            100 * sum(specific_genes %in% names(gene2GO)) / length(specific_genes)))
```

With GO term assigned: 17 (7.5%)

```
cat("\n")
```

```
cat("Significant GO BP terms:           ", sum(go_sig$Ontology == "BP"), "\n")
```

Significant GO BP terms: 6

```
cat("Significant GO MF terms:           ", sum(go_sig$Ontology == "MF"), "\n")
```

Significant GO MF terms: 6

```
cat("Significant GO CC terms:           ", sum(go_sig$Ontology == "CC"), "\n")
```

Significant GO CC terms: 2

```
cat("Significant Pfam domains:          ", nrow(pfam_sig), "\n")
```

Significant Pfam domains: 5

11 10. Session info

```
sessionInfo()
```

R version 4.3.1 (2023-06-16)

Platform: aarch64-apple-darwin20 (64-bit)

Running under: macOS 26.3.1

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRblas.0.dylib

LAPACK: /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/lib/libRlapack.dylib; LAPACK

locale:

[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Stockholm

tzcode source: internal

attached base packages:

[1] stats4 stats graphics grDevices utils datasets methods

[8] base

other attached packages:

[1] scales_1.4.0	topGO_2.54.0	SparseM_1.84-2
[4] GO.db_3.18.0	AnnotationDbi_1.64.1	IRanges_2.36.0
[7] S4Vectors_0.40.2	Biobase_2.62.0	graph_1.80.0
[10] BiocGenerics_0.48.1	data.table_1.18.2.1	lubridate_1.9.4
[13] forcats_1.0.1	stringr_1.6.0	dplyr_1.1.4
[16] purrr_1.2.2	readr_2.1.5	tidyr_1.3.1
[19] tibble_3.3.1	ggplot2_4.0.3	tidyverse_2.0.0

loaded via a namespace (and not attached):

[1] KEGGREST_1.42.0	gtable_0.3.6.9000	xfun_0.57
[4] lattice_0.22-9	tzdb_0.5.0	bitops_1.0-9
[7] vctrs_0.7.3	tools_4.3.1	generics_0.1.4
[10] parallel_4.3.1	RSQLite_2.4.1	blob_1.3.0
[13] pkgconfig_2.0.3	RColorBrewer_1.1-3	S7_0.2.2
[16] GenomeInfoDbData_1.2.11	lifecycle_1.0.5	compiler_4.3.1
[19] farver_2.1.2	textshaping_1.0.1	Biostrings_2.70.3
[22] GenomeInfoDb_1.38.8	htmltools_0.5.9	RCurl_1.98-1.18
[25] yaml_2.3.12	pillar_1.11.1	crayon_1.5.3
[28] cachem_1.1.0	tidyselect_1.2.1	digest_0.6.37
[31] stringi_1.8.7	labeling_0.4.3	fastmap_1.2.0
[34] grid_4.3.1	cli_3.6.5	magrittr_2.0.5
[37] utf8_1.2.6	dichromat_2.0-0.1	withr_3.0.2
[40] bit64_4.8.0	timechange_0.3.0	rmarkdown_2.31
[43] XVector_0.42.0	httr_1.4.8	matrixStats_1.5.0
[46] bit_4.6.0	otel_0.2.0	ragg_1.4.0
[49] png_0.1-9	hms_1.1.4	memoise_2.0.1
[52] evaluate_1.0.5	knitr_1.51	viridisLite_0.4.3
[55] rlang_1.2.0	glue_1.8.1	DBI_1.3.0
[58] vroom_1.6.5	rstudioapi_0.18.0	jsonlite_2.0.0
[61] R6_2.6.1	systemfonts_1.2.3	zlibbioc_1.48.2